

Mathematics Bootcamp

Optimization

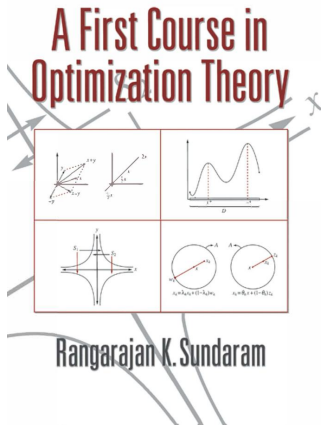
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Today

- Existence of a Maximum
- First- and Second-Order Conditions
- Concavity and Uniqueness
- Equality Constraints and Lagrange
- Inequality Constraints and Karush–Kuhn–Tucker
- The Consumer Problem

Sundaram - Optimization Theory



- Most of what follows is taken from Sundaram.
- Written for economists: existence before characterization.
 - Every theorem is stated for a purpose.
 - Proofs kept separate from use.

Notation

Fixed Throughout

$$D \subseteq \mathbb{R}^n, \quad f : D \rightarrow \mathbb{R}, \quad \max\{f(x) \mid x \in D\}.$$

One dimension: $f'(x)$, $f''(x)$.

Several dimensions, the **gradient** and the **Hessian**:

$$\nabla f(x) := \begin{pmatrix} f_1(x) \\ \vdots \\ f_n(x) \end{pmatrix}, \quad H_f(x) := \begin{pmatrix} f_{11}(x) & \cdots & f_{1n}(x) \\ \vdots & \ddots & \vdots \\ f_{n1}(x) & \cdots & f_{nn}(x) \end{pmatrix}.$$

$$f_i = \partial f / \partial x_i \text{ and } f_{ij} = \partial^2 f / \partial x_i \partial x_j.$$

The Optimization Problem

Sundaram 2.1

Given $f : \mathbb{R}^n \rightarrow \mathbb{R}$ and $D \subseteq \mathbb{R}^n$,

$$\max\{f(x) \mid x \in D\} \quad \text{or} \quad \min\{f(x) \mid x \in D\}.$$

f is the **objective**, D the **feasible set**.

Two questions: does a solution exist, and how do we find it?

Maximum Value or Maximizing Choice

Two Different Objects

$$\max_{x \in D} f(x) \in \mathbb{R}, \quad \arg \max_{x \in D} f(x) = \{x \in D : f(x) \geq f(y) \forall y \in D\}.$$

The first is a number; the second is a set, possibly empty.

Increasing Transformations

Proposition

If $\phi : \mathbb{R} \rightarrow \mathbb{R}$ is strictly increasing, then

$$\arg \max_{x \in D} f(x) = \arg \max_{x \in D} (\phi \circ f)(x).$$

Utility is ordinal (Day 2): the objective may be rescaled freely.

Existence: The Weierstrass Theorem

Sundaram 3.1

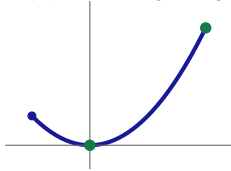
Let $D \subseteq \mathbb{R}^n$ be **compact**, and let $f : D \rightarrow \mathbb{R}$ be **continuous** on D . Then f attains a maximum and a minimum on D .

$$D \text{ compact, } f \text{ continuous} \implies \arg \max_{x \in D} f(x) \neq \emptyset.$$

Compact means closed and bounded (Day 2, Heine–Borel).

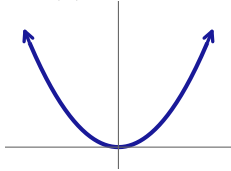
Dropping One Assumption

$$f(x) = x^2 \text{ on } [-1, 2]$$



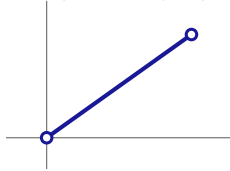
D compact, f continuous
 $\Rightarrow$ max and min attained

$$f(x) = x^2 \text{ on } \mathbb{R}$$



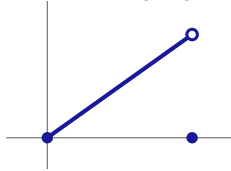
D not bounded
 $\Rightarrow$ no maximum

$$f(x) = x \text{ on } (0, 1)$$



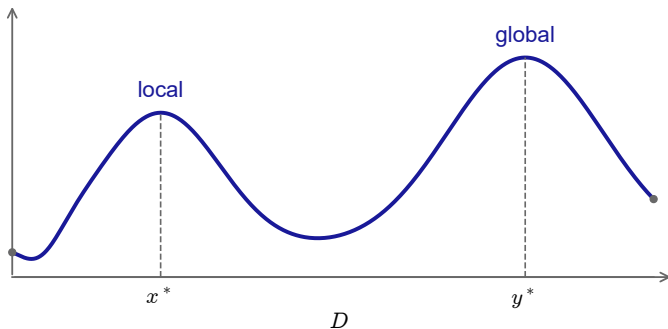
D not closed
 $\Rightarrow$ neither attained

$$\text{jump on } [0, 1]$$



f not continuous
 $\Rightarrow$ no maximum

Global and Local Maxima



A local maximum beats its neighbors; a global maximum beats everything.

The First-Order Condition

Sundaram 4.1

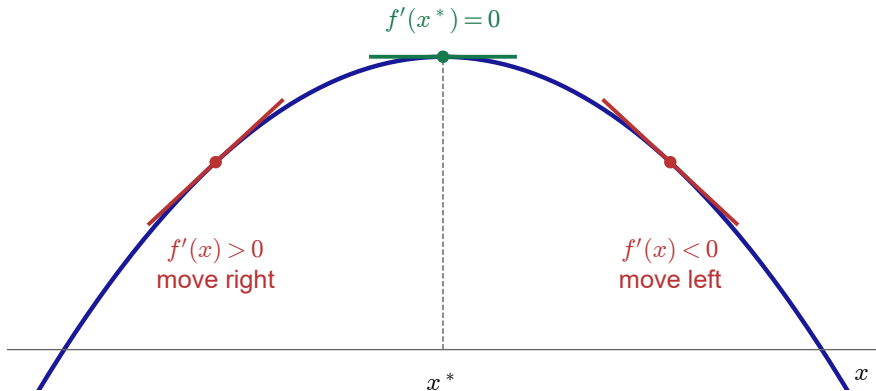
Suppose $x^* \in \text{int } D \subseteq \mathbb{R}^n$ is a local maximum of f on D , and f is differentiable at x^* . Then

$$\nabla f(x^*) = 0.$$

The same holds if x^* is a local minimum.

Interior is an assumption: at a boundary point the condition fails.

A Nonzero Slope

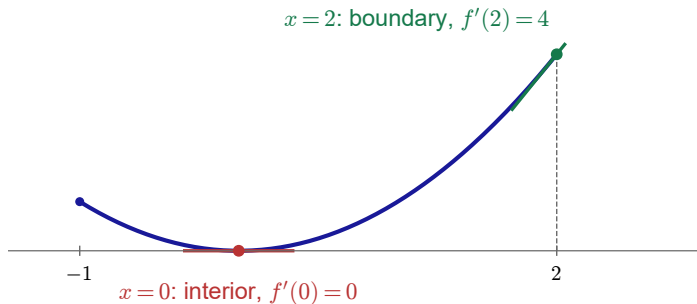


A nonzero slope leaves a feasible direction of improvement.

Stationary Points and the Boundary

Osborne 5.1.1

Let $f(x) = x^2$ on $D = [-1, 2]$. The only stationary point, $x = 0$, is the global minimum; the global maximum is the boundary point $x = 2$, where $f'(2) \neq 0$.

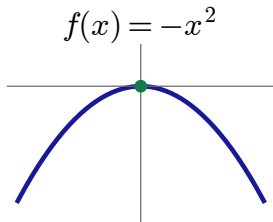


Solve the first-order condition, then compare against the boundary.

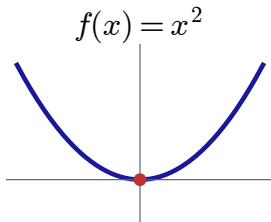
Critical Points

Critical Point

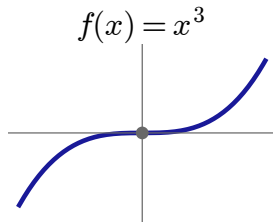
x^* is a **critical point** of f on D if $\nabla f(x^*) = 0$.



local and global
maximum



local and global
minimum



neither

The first-order condition is necessary, never sufficient (Sundaram 4.2).

Second-Order Conditions

Sundaram 4.3

Let f be C^2 and let $x^* \in \text{int } D$ be a critical point.

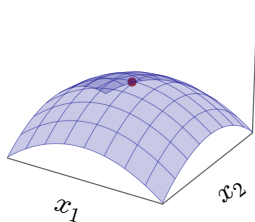
$$x^* \text{ local maximum} \implies H_f(x^*) \preceq 0,$$

$$\nabla f(x^*) = 0, H_f(x^*) \prec 0 \implies x^* \text{ strict local maximum.}$$

Semidefinite is necessary; definite is sufficient.

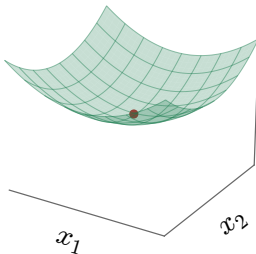
The Hessian

negative definite



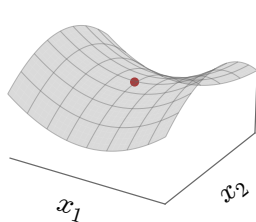
strict local maximum

positive definite



strict local minimum

indefinite



neither

Locally concave, locally convex, neither.

The Determinant Test in $\mathbb{R}^2$

Proposition

For $H = H_f(x^*)$ with entries f_{11}, f_{12}, f_{22} ,

$f_{11} < 0, \det H > 0 \implies$ local maximum,

$f_{11} > 0, \det H > 0 \implies$ local minimum,

$\det H < 0 \implies$ saddle.

Definiteness and Concavity

Sundaram 7.10

Let $D \subseteq \mathbb{R}^n$ be open and convex and let f be C^2 . Then

$$f \text{ concave on } D \iff H_f(x) \preceq 0 \text{ for every } x \in D,$$

and $H_f(x) \prec 0$ for every $x \in D$ implies f strictly concave.

At one point definiteness is a local test; at every point it is concavity.

Equality Constraints

Sundaram 5.1

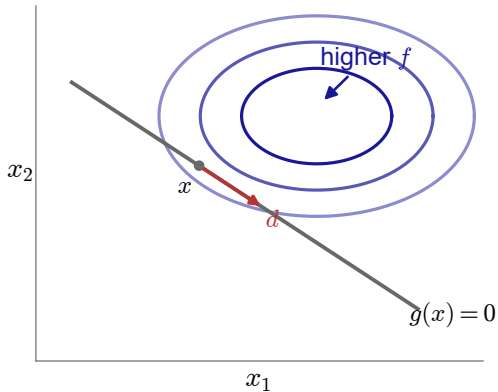
For $f : \mathbb{R}^n \rightarrow \mathbb{R}$ and $g : \mathbb{R}^n \rightarrow \mathbb{R}^k$,

$$\max f(x) \quad \text{subject to} \quad g_i(x) = 0, \quad i = 1, \dots, k.$$

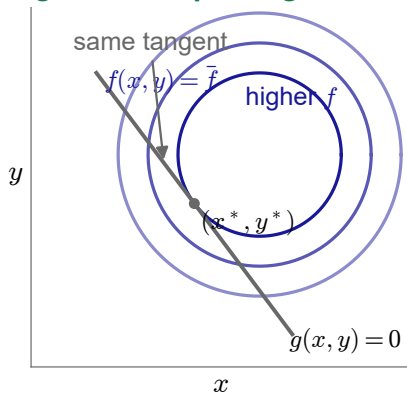
The constraint removes k degrees of freedom.

Tangency at an Optimum

crossing: improve along g



tangent: no improving direction



The Theorem of Lagrange

Sundaram 5.1

Let f and g_i be C^1 , and let x^* be a local maximum or minimum of f on

$$D = U \cap \{x \mid g_i(x) = 0, i = 1, \dots, k\}, \quad U \text{ open.}$$

If $\nabla g_1(x^*), \dots, \nabla g_k(x^*)$ are linearly independent, there is $\lambda^* \in \mathbb{R}^k$ with

$$\nabla f(x^*) + \sum_{i=1}^k \lambda_i^* \nabla g_i(x^*) = 0.$$

Lagrange in Two Variables

Osborne 6.1

Maximize $f(x, y) = xy$ subject to $x + y = 6$. With $L(x, y) = xy - \lambda(x + y - 6)$ the conditions

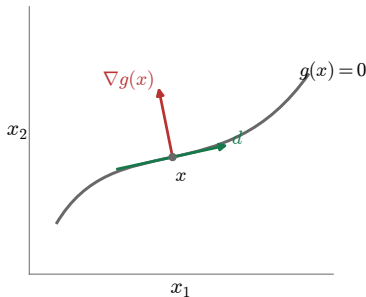
$$y - \lambda = 0, \quad x - \lambda = 0, \quad x + y = 6$$

give $(x^*, y^*) = (3, 3)$ and $\lambda^* = 3$.

Regularity

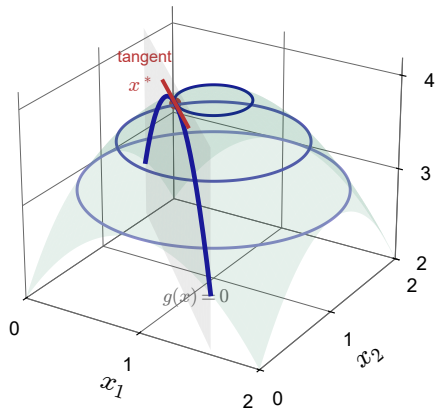
Constraint Qualification

The rank condition $\nabla g_1(x^*), \dots, \nabla g_k(x^*)$ are linearly independent says the constraint gradients are linearly independent at x^* .



Lagrange gives necessary conditions only, and only where this rank condition holds.

The Constraint Cuts a Curve Out of the Surface



The optimum is the highest point of that curve.

The Cobb–Douglas Consumer

Osborne 6.1

Maximize $x^a y^b$ subject to $px + y = m$, with $a, b, p, m > 0$. From

$$ax^{a-1}y^b = \lambda p, \quad bx^a y^{b-1} = \lambda, \quad px + y = m$$

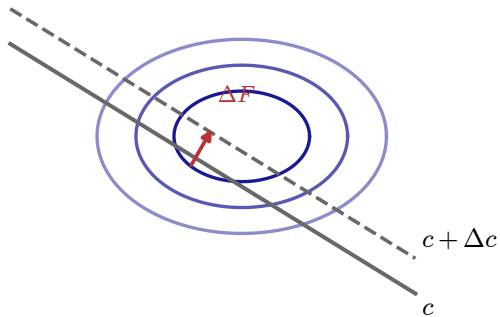
we get

$$x^* = \frac{am}{(a+b)p}, \quad y^* = \frac{bm}{a+b}.$$

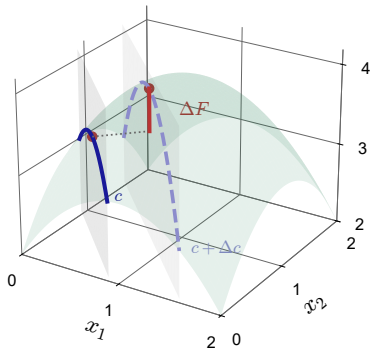
Fixed income shares, independent of p and m .

The Multiplier as a Shadow Value

Level sets



Objective surface



Relaxing the constraint from c to $c + \Delta c$ raises the optimum by $\Delta F \approx \lambda^* \Delta c$.

Inequality Constraints

Sundaram 6.1

For $f : \mathbb{R}^n \rightarrow \mathbb{R}$ and $h : \mathbb{R}^n \rightarrow \mathbb{R}^l$,

$$\max f(x) \quad \text{subject to} \quad h_i(x) \geq 0, \quad i = 1, \dots, l.$$

A constraint is **effective** at x^* if $h_i(x^*) = 0$.

Which constraints bind is not given in advance; it is part of the solution.

The Karush–Kuhn–Tucker Theorem

Sundaram 6.1

Let f and h_i be C^1 , and let x^* be a local maximum of f on

$$D = U \cap \{x \mid h_i(x) \geq 0, i = 1, \dots, l\}, \quad U \text{ open.}$$

If the effective gradients $\{\nabla h_i(x^*)\}_{i \in E}$ are linearly independent, there is $\lambda^* \in \mathbb{R}^l$ with

$$\lambda_i^* \geq 0, \quad \lambda_i^* h_i(x^*) = 0, \quad \nabla f(x^*) + \sum_{i=1}^l \lambda_i^* \nabla h_i(x^*) = 0.$$

Complementary Slackness

Two Regimes

For each constraint, $\lambda_i^* h_i(x^*) = 0$ forces one of

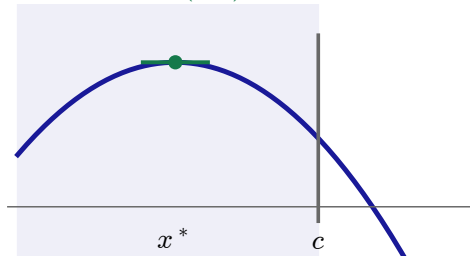
$h_i(x^*) = 0$: the constraint binds, and $\lambda_i^* \geq 0$;

$h_i(x^*) > 0$: the constraint is slack, and $\lambda_i^* = 0$.

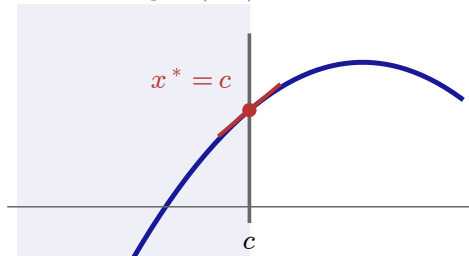
A slack constraint has no shadow value.

Complementary Slackness

slack: $h(x^*) > 0, \lambda^* = 0$



binding: $h(x^*) = 0, \lambda^* > 0$



Maximize f subject to $h(x) = c - x \geq 0$.

From Local to Global

Sundaram 7

Let D be convex and f concave on D . Then every local maximum of f is a global maximum, and the set of maximizers is convex.

For concave f the first-order condition is also sufficient.

Uniqueness

Sundaram 8

Let D be convex and f *strictly* quasiconcave. Then f has at most one maximizer on D .

Day 3 promised this: strictly convex preferences give a single demand.

A Binding Budget

Proposition

If $\succeq$ is locally nonsatiated and $p \gg 0$, every solution of

$$\max u(x) \quad \text{subject to} \quad p \cdot x \leq I$$

satisfies $p \cdot x = I$.

Day 3 promised this too: the inequality becomes an equality.

The Consumer Problem

Utility Maximization

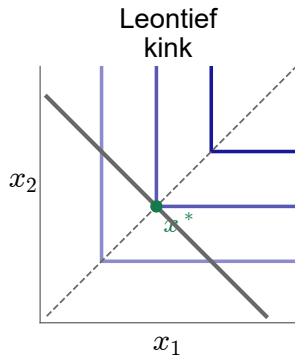
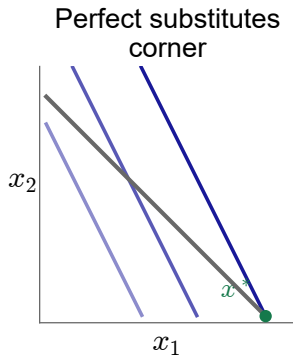
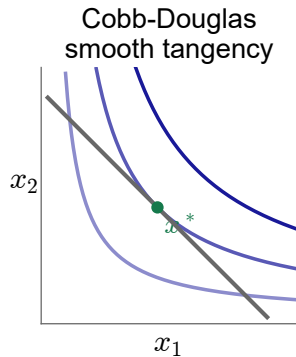
Given prices $p \gg 0$ and income $I > 0$,

$$\max_{x \in \mathbb{R}_+^L} u(x) \quad \text{subject to} \quad p \cdot x \leq I,$$

over the budget set $B(p, I) = \{x \in \mathbb{R}_+^L : p \cdot x \leq I\}$, which is compact and convex.

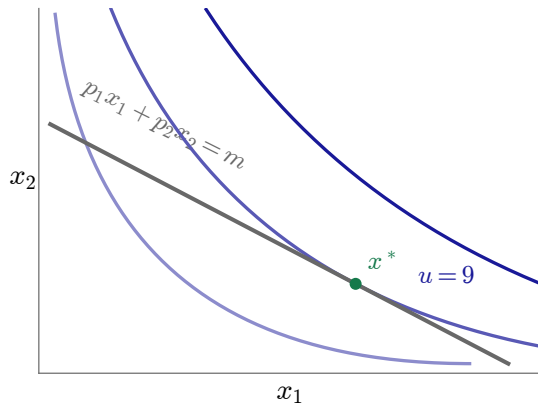
Compact and continuous give existence; strictly convex preferences give uniqueness.

Three Utility Forms, Three Methods



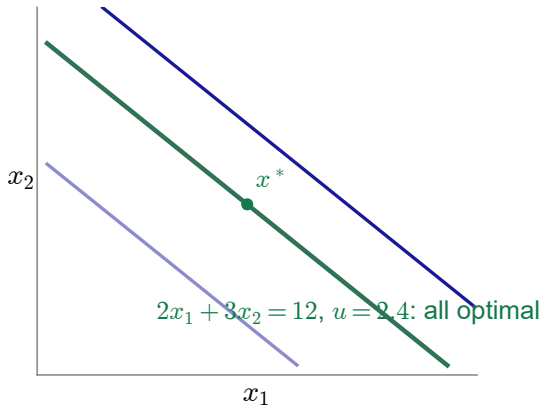
An interior tangency, a corner, and a kink.

CES: An Interior Solution



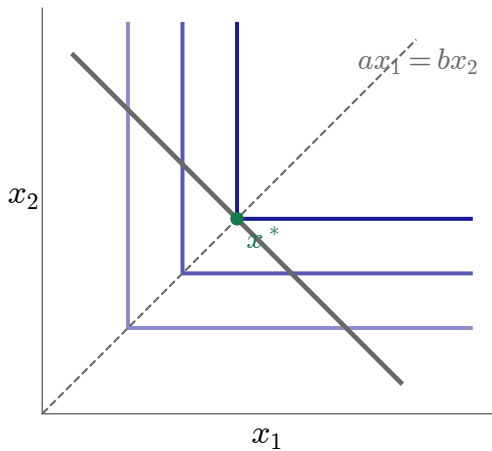
Smooth and strictly convex: the tangency condition solves it.

Perfect Substitutes: A Corner



No tangency exists: the solution is a corner unless the ratios coincide.

Leontief: The Kink



u is not differentiable at the optimum: use the constraint directly.

Conditions and Conclusions

Assumption	Conclusion
Compact D , continuous f	a maximum exists
Interior optimum, f differentiable	$\nabla f(x^*) = 0$
$H_f(x^*) \prec 0$	local maximum
D convex, f concave	local becomes global
f strictly quasiconcave	at most one maximizer
Constraint gradients independent	multipliers exist

Day 5 puts these to work over an infinite horizon.